

TABLE I: Related Work

Ref	Relation	Year	Categories	Name	Brief Abstract	Algorithm	Performance Metrics	Experiment Setup/Simulation	Comparison
[1]	Near	2021	OMR, NLP, Automation	Generation and Grading of Arduous MCQs using NLP and OMR Detection using OpenCV	Integrates NLP-based MCQ generation with OpenCV OMR grading. Handles multi/partial answers using thresholding and contour detection.	PEGASUS transformer + Naïve Bayes + OpenCV (Otsu threshold, Canny edge).	Visual results show accurate bubble detection; no numeric accuracy reported.	Synthetic OMR sheets simulated in Python/OpenCV.	Comparable to manual grading and conventional OMR scanners.
[2]	Near	2024	OMR, Deep Learning, Review	Optical Mark Recognition Techniques for Multiple-Choice Tests: A Comprehensive Review	Comprehensive review of 25 OMR studies from traditional image processing to modern deep learning approaches.	Comparative analysis across thresholding, Hough transform, CNN, and YOLO-based models.	Reported accuracies range from 97.3%–99.98%; YOLO-based models achieve <0.5% error, pixel-counting 99.94%.	Analysis includes traditional, CV, and DL-based simulations; runtime up to 50 sheets/min.	Summarizes methods' strengths, weaknesses, and future research gaps in OMR.
[3]	Direct	2025	OMR, Image Processing, Large-Scale Evaluation	Unsupervised Optical Mark Recognition on Answer Sheets for Massive Printed Multiple-Choice Tests	Presents an unsupervised, parameter-free OMR algorithm for large-scale exam evaluation using 6029 real answer sheets.	Adaptive thresholding, contour detection, morphological filtering, and multi-stage correction with alternative processes.	Achieved 96.15% of exams graded without error and 99.95% of individual answers correctly classified; average time 1.05 s/sheet.	Tested on 6029 real scanned answer sheets (564,040 responses) from 44 high schools in Mexico.	Compared with Circle Hough Transform, outperforming it (99.26% vs 80.64% circle detection accuracy).
[4]	Direct	2025	AI, OCR, Education, Vietnam Exam Reform	Application of AI in Grading New Question Types in the 2025 Vietnamese High School Graduation Exam	Develops an AI-based grading system combining Mathpix OCR with C for recognizing handwritten responses on redesigned answer sheets supporting MCQ, True/False, and short-answer types.	Mathpix OCR handwriting recognition integrated in a C environment using a five-stage Borg	Gall model.	Field-tested in two classrooms with 93 students and 38 teachers.	Outperformed traditional OMR scanners (97–98%) and YOLOv8-based models lacking short-answer support.
[5]	Direct	2021	OMR, Image Processing, Template-Free	Image Processing for Enhanced OMR Answer Matching Precision	Proposes a flexible OMR system that works across multiple template designs without predefined layouts.	OpenCV-based pipeline using Gaussian blur, Otsu thresholding, contour detection, dilation, and ROI extraction.	Achieved 100% F1-score for 50–90 question templates and 99% accuracy for 120 questions (avg error 0.668%).	Tested on 3 OMR templates (50, 90, 120 questions) × 5 samples each using scanned images.	Outperforms prior template-dependent methods (up to 90 questions) in accuracy and flexibility.
[6]	Direct	2019	OMR, Machine Learning, CNN	The Achievement of Higher Flexibility in Multiple-Choice-Based Tests Using Image Classification Techniques	Introduces a machine learning-based OMR system capable of distinguishing confirmed, crossed-out, and empty answers using real exam data.	Naïve Bayes, Bag of Visual Words (BoVW), and fine-tuned AlexNet CNN with two-stage classification strategy.	CNN: 99.78% question accuracy, 97.69% sheet accuracy; cross-device generalization 93.3% (scanner) and 90.9% (mobile).	Dataset of 735 answer sheets (6 exams, 33,549 ROIs); experiments on CPU/GPU setups using MATLAB.	Outperforms thresholding and earlier ML OMR systems by 44% in grading accuracy and robustness.
[7]	Direct	2020	OMR, Security, Cryptography	A Novel Approach for the Detection of OMR Sheet Tampering Using Encrypted QR Code	Introduces an encrypted QR-code mechanism to prevent and detect post-exam tampering on OMR sheets.	Elliptic Curve Cryptography (ECC) + ZXing QR-code encoding/decoding.	Detected all 6 tampered sheets (100% detection) and correctly verified 4 genuine ones.	Experiment on 10 OMR sheets using two ECC key-pairs (institution / evaluation center).	Compared with standard OMR systems lacking tampering detection capability.
[8]	Direct	2019	OMR, Image Processing, Damage Recovery	Information Retrieval From Damaged OMR Sheet by Using Image Processing	Proposes a low-cost scanner-based OMR system that recovers information from damaged or torn answer sheets rejected by traditional OMR machines.	Circular Hough Transform (CHT) for bubble detection, orientation correction, and template matching using MATLAB/Python.	Achieved 97.07% localization accuracy and 97.80% recognition accuracy for damaged OMR sheets.	Tested on scanned answer sheets from Bangladesh Board exams using normal scanners.	Outperforms commercial OMR machines which cannot process damaged forms.
[9]	Direct	2019	OMR, Machine Learning, Classification	Contrasting Classifiers for Software-based OMR Responses	Compares four classifiers (pixel count, pattern-based, bit-string, and edge detection) to identify selected and deselected answers on software-based OMR sheets.	Threshold-based pixel count, decision tree on 9×9 pattern values, Levenshtein edit distance, and AForge.NET edge detection.	Pixel-count classifier achieved 99.98% accuracy, 1.0 sensitivity, 0.9998 specificity, K = 0.9995; edge-based classifier 73.6% accuracy, but 45% success on deselected answers.	Datasets: 74 training and 60 test sheets (total 35,895 hotspots), plus 20 validation hotspots for deselection detection.	Outperformed complex ML classifiers for general OMR; edge-based uniquely identified deselections.

Continued on next page

Table I (continued)

Ref	Relation	Year	Categories	Name	Brief Abstract	Algorithm	Performance Metrics	Simulation	Comparison
[10]	Near	2021	Steganography, OMR, Machine Learning	Coverless Image Steganography Based on Optical Mark Recognition and Machine Learning	Proposes a robust coverless steganography method using OMR bubble sheets as cover carriers instead of digital images.	Rule-Based Machine Learning (RBML) + OMR detection algorithms for binary bit mapping and extraction.	Achieved 100% robustness (BER = 0) against JPEG compression, noise (0.01–0.04), thresholding, color conversion, and print-scan attacks.	Simulated bubble-sheet templates (4×30 layout) using Microsoft Word and OMR fonts; tested under multiple attack scenarios.	Outperforms state-of-the-art coverless steganography in robustness, security, and capacity (120 bits/sheet, 68 images for 1 KB data).
[11]	Near	2024	OMR, Education, Software Evaluation	Optical Mark Recognition in Student Continuous Assessment	Compares open-source (AMC) and low-cost (GEXCAT) OMR tools for continuous assessment in university courses.	OMR-based test generation and grading integrated with GEECA using LaTeX macros and AMC/GEXCAT pipelines.	AMC: 98% recognition rate (2% student-error); GEXCAT: >95% valid scans after correction.	Tested with 630 students (≈3150 sheets) across two UPM schools using regular scanners (300 dpi).	Both tools reduced grading time > 70%; AMC preferred for open-questions, GEXCAT for MCQs.
[12]	Direct	2023	OMR, Mobile Application, Image Processing	Mobile Based MCQ Answer Sheet Analysis and Evaluation Application	Proposes an Android app for analyzing and grading MCQ answer sheets without OMR machines or special paper.	Adaptive thresholding, pixel-count method, ROI cropping, and contour-based detection implemented in Android using OpenCV.	Average accuracy 99.44% at 45% pixel threshold, 98.78% at 42%; optimal threshold = 45%.	Tested on 5 MCQ sheets (12–120 questions) captured via smartphone camera.	Outperforms manual checking and maintains >97% accuracy under variable lighting.
[13]	Direct	2018	OMR, Template Matching, Android	Automated Scoring of Multiple-Choice Test Using Template Matching Technique	Develops a low-cost Android-based scoring system using X-mark detection instead of filled circles.	Template matching with normalization (11×11 mask) and affine transformation for X-mark recognition.	99% success rate; 0.73% miss rate for corrected X-marks; overall error ≈0.01% on 14 answer sheets.	Evaluated on 27 + 14 answer sheets, 1820 X-mark, and 1560 corrected X-mark images.	Outperforms previous methods by simplifying detection while supporting answer correction.
[14]	Direct	2019	OMR, Parameter-Less, Image Processing	Automated Parameter-Less Optical Mark Recognition	Proposes a robust, template-independent OMR algorithm that dynamically localizes mark regions without prior layout information.	Adaptive local thresholding, morphological closing, contour grouping, and forward-backward search to recover missed regions.	Achieved <2% error rate; processing 150 sheets per minute in real time.	Evaluated on multiple OMR templates captured from scanners and smartphones with varying circle sizes and layouts.	Outperforms template-based methods, enabling real-time, low-cost, device-independent OMR grading.
[15]	Direct	2019	OMR, Image Processing, Large-Scale Testing	Optical Mark Recognition Based on Image Processing Techniques for the Answer Sheets of the Colombian High-Stakes Tests	Presents a large-scale OMR algorithm using dual segmentation and centroid-based mark recognition for national exams (Saber 11, Colombia).	Preprocessing with Otsu thresholding and morphological hole filling; two-step segmentation (bubble block extraction and mark recognition).	Calibration: 99.83% accuracy on 1600 images; Large-scale test: 99.7% accuracy on 64,940 sheets; avg 0.1 s/sheet on 8-core PC.	Evaluated on 65k scanned sheets (Saber 11 exam, 2018-2) using parallel multiprocessing in Python.	Outperforms traditional OMR scanners in cost and throughput (8 sheets/s, 99.7% accuracy).
[16]	Near	2021	OMR, Review, Benchmark	Optical Mark Recognition: Advances, Difficulties, and Limitations	Comprehensive review of 35 OMR papers covering techniques, datasets, processing times, and restrictions in hardware- and software-based OMR.	Comparative survey of pixel counting, Hough transform, template matching, machine learning (NB, QDF, CNN), and fiducial-marker-free methods.	Average accuracy 98.85%, error 1.08%; avg processing time 2.78 s per sheet; best 0.34 s (parallel), worst 68 s.	Analyzes 35 works with reported datasets (5 public); compiles restrictions (A–J) and generations (1st–4th).	Defines 4th gen OMR as flexible template-free software achieving > 99 % accuracy and ≤ 3s per sheet.
[17]	Direct	2019	OMR, Alignment, Skew Correction	Alignment, Scale and Skew Correction for Optical Mark Recognition Documents Based	Proposes a document-alignment algorithm correcting translation, rotation, and scaling distortions in OMR answer sheets using pattern matching on ROI columns.	Binarization, morphological filtering (AForge.NET), heuristic column grouping, and quadrilateral transformation via homogeneous coordinates.	Experiment 1: 100% success (40 synthetic images); Experiment 2: 98.75% success (1035 real tests, 1.25% error).	Evaluated on 40 synthetically distorted and 1035 real student tests (240 dpi).	Simpler and faster than Fourier–Mellin and Hough-based corrections; robust for large document batches.
[18]	Direct	2022	OMR, Optimization	A NOVEL OPTICAL MARK RECOGNITION TECHNIQUE BASED ON BIOGEOGRAPHY-BASED OPTIMIZATION	Introduces a BBO-based optimization approach for OMR recognition to improve robustness.	Neocognitron (early CNN).	Recognition accuracy.	Not specified.	Not specified.
[19]	Near	2022	OMR, Low-cost	Cost effective optical mark recognition software for educational institutions	Develops a robust, low-cost OMR system for automated data entry with support for varying image conditions.	ML classifiers; CNN.	Accuracy.	Not specified.	Compared with expensive OMR systems.

Continued on next page

Table I (continued)

Ref	Relation	Year	Categories	Name	Brief Abstract	Algorithm	Performance Metrics	Simulation	Comparison
[20]	Direct	2023	OMR, Webcam	Webcam Based Real-Time Robust Optical Mark Reader for MCQ Test Evaluation	Real-time OMR using a webcam for capturing and grading.	Real-time capture and processing.	Accuracy; latency.	Not specified.	Not specified.
[21]	Near	2020	OMR Hardware, FPGA	A high-speed optical mark reader hardware based on programmable logic for low-cost applications	Implements a high-speed, low-cost OMR reader on programmable logic.	Hardware OMR pipeline.	Throughput (e.g., frames/s).	Hardware prototype.	Compared with software-based solutions.
[22]	Direct	2023	OMR, Architecture	An Image Processing Oriented Optical Mark Reader Based on a Modified Multi-Connect Architecture	Presents an image-processing oriented OMR reader based on a modified multi-connect architecture.	Modified multi-connect architecture.	Accuracy; speed.	Not specified.	Compared with academic OMR works.
[23]	Direct	2021	OMR, Camera-based	Automatic Correction of Multiple-Choice Tests Using Hough Transform and Neural Network	Camera-based OMR correction with detection and deskewing followed by normalization and recognition.	Hough transform; normalization.	Accuracy; processing time.	Not specified.	Compared with traditional OMR.
[24]	Near	2019	OMR, Usability	Automated marking of printed multiple choice answer sheets	Focuses on user-friendly OMR form design and interaction aspects.	Form design guidelines.	Usability indicators.	Not specified.	Not applicable.
[25]	Near	2020	Education, Learning	Does a Distributed Practice Strategy for Multiple Choice Questions Help Novices Learn Programming?	Educational study on practice strategies with MCQs; not an OMR algorithm.	Experimental study design.	Learning outcomes.	Not applicable.	Comparison between practice strategies.

Table 2 describes in detail the dataset and self-assessment opinions about related works (because Table 1 has enough columns, these fields are moved to Table 2).

TABLE II: Related Work Detail

Ref	Dataset	Comment	Note
[1]	Synthetic MCQ and OMR sheets generated via NLP; no public dataset used.	Combines PEGASUS-based question generation with OpenCV grading; detects partial and multiple answers effectively.	Qualitative results show correct bubble detection and scoring; no quantitative accuracy reported.
[2]	Reviewed 25 datasets across OMR research; includes both scanner-based and camera-based answer sheets; no unified public dataset available.	Summarizes accuracy, error rate, recall, and speed metrics used in OMR evaluation; highlights gap in benchmark datasets.	Quantitative results: accuracies up to 99.98%, recall 99.03%, runtime 0.12–1.2 s/sheet depending on method.
[3]	Dataset of 6029 scanned answer sheets (564,040 four-option responses) collected from 44 Mexican high schools; publicly unavailable.	Implements an unsupervised algorithm robust to rotation, illumination, multiple answers, and partial marks; includes a user interface for batch grading.	Results: 96.15% sheets graded without error, 99.95% correct responses, 1.05 s/sheet (≈ 57 sheets/min), two orders of magnitude faster than manual grading.
[4]	The dataset includes 93 real exam papers from Grade-12 students and 38 teachers across Thai Nguyen province, used for AI validation and feedback.	System integrates direct-write answer sheets and Mathpix OCR; recognized A–D, T/F, 0–9 entries; improved speed and reduced shading errors.	Results: 98.5% recognition accuracy, teacher satisfaction $\geq 4.0/5$ Likert, processing time shorter than OMR; limitations include handwriting variability and 300 dpi hardware needs.
[5]	Dataset of 3 OMR template types (50, 90, 120 questions) $\times$ 6 sheets each; images captured via phone camera and processed in Python (OpenCV).	Implements fully template-free OMR marking through contour-based ROI detection and adaptive thresholding; consistent results across different form designs.	Results: F1 score = 100% (≤ 90 questions), 99% (120 questions); average error 0.668%; processing time ≈ 1 s per sheet.
[6]	Dataset: 735 answer sheets from six real MCQ exams (10,980 confirmed, 202 crossed-out, 22,367 empty); publicly available PNG/XPS format.	Proposes ML-based OMR with Naïve Bayes, BoVW (SVM), and CNN; introduces two-stage classification for improved robustness across forms and devices.	Results: CNN 99.78% (question), 97.69% (sheet); BoVW 92.3% $\rightarrow$ 99.26% (two-stage); cross-device generalization 93.3% (scanner), 90.9% (mobile); time ≈ 0.5 –1.6 s per question.
[7]	Dataset of 10 OMR sheets (6 tampered + 4 genuine) processed via custom encoding and detection modules using ECC + ZXing.	Encrypts OMR metadata (total responses, sheet ID, enrollment ID) into a QR code; decodes and verifies consistency at evaluation center.	Results: 100% detection of tampered sheets; 0 false positives on genuine ones; efficiency confirmed through comparison table.
[8]	Dataset from Bangladesh Education Board exams; includes multiple damaged, folded, and stained OMR sheets scanned via regular flatbed scanners.	Employs CHT-based mark detection with preprocessing (binarization, rotation, template alignment) to extract answers from corrupted regions.	Results: 97.07% localization accuracy, 97.80% recognition accuracy, lower cost ($\approx 10\times$ cheaper) than OMR hardware, and 20–40% slower processing time.
[9]	Dataset includes 74 training and 60 test OMR sheets (35,895 hotspots) with 20 deselected entries for validation.	Evaluates four classification strategies using R (rpart) and AForge.NET; edge-count and pixel-density features extracted from 72 $\times$ 72 hotspots.	Results: Pixel-count 99.98% accuracy, 1.0 sensitivity, 0.9998 specificity, K = 0.9995; edge-based correctly detected 45% deselections—unique among tested methods.
[10]	Dataset: synthetic OMR bubble sheets (4 $\times$ 30) generated via OMR font; no real student data used; evaluated under multiple attack types.	Combines RBML mapping and OMR detection to encode binary payloads as marked bubbles, eliminating traditional cover databases.	Results: 100% BER robustness for JPEG compression, noise ≤ 0.04 , scaling (0.5–1.5), color conversion, and print-scan; embedding capacity = 120 bits/sheet, highest among reviewed methods.
[11]	Dataset of 3150 OMR sheets from 630 students (two UPM engineering schools) scanned at 300 dpi B/W.	Evaluates AMC (GNU/Linux + LaTeX) and GEXCAT (Windows) within GEECA framework for continuous assessment. Both reduced grading time and teacher workload while maintaining accuracy.	Results: AMC 98% recognition success (2% student errors); GEXCAT $>95\%$ after correction; grading time cut $>70\%$; user satisfaction high.
[12]	Dataset: 5 answer sheets (12–120 questions each) photographed under varying lighting; processed through Android app with OpenCV backend.	Implements pixel-density counting after adaptive thresholding; determines marked answers based on 40–48% black-pixel ratio.	Results: Avg accuracy 99.44% (45% threshold); 98.78% (42% threshold); best trade-off at 45%; low cost and mobile-friendly OMR alternative.
[13]	Dataset includes 27 + 14 answer sheets, 1820 X-mark and 1560 corrected X-mark samples; captured via mobile camera under variable angles.	Implements 4-stage pipeline: image prep $\rightarrow$ blank-square detection $\rightarrow$ X-mark detection $\rightarrow$ scoring; uses S2M2 mask achieving best recognition accuracy.	Results: 99% overall detection, 0.73% missed corrected marks, error rate $\approx 0.01\%$; simple, fast, supports corrected answers unlike circle-based OMR.

Continued on next page

Table II (continued)

Ref	Dataset	Comment	Note
[14]	Dataset includes OMR sheets with varying templates, scanner resolutions, and smartphone captures; total 150 sheets/min throughput benchmark.	Implements fully parameter-less mark localization with dynamic ROI detection, local threshold computation per row, and heuristics for missed marks.	Results: <2% total error rate, real-time speed (150 sheets/min), robust to tilt, noise, poor markings, and scribbles; efficient on standard PCs without GPUs.
[15]	Dataset of 64,940 scanned answer sheets from Saber 11 2018-2 national exams (Colombia); calibration set: 1600 images, test set: 65k.	Implements two-stage segmentation (bubble block extraction, mark recognition) using Otsu thresholding and centroid matching; parallel processing across 8 logical cores.	Results: 99.83% calibration accuracy, 99.7% large-scale accuracy, 0.1 s/sheet (≈ 8 sheets/s); type-I errors mainly due to scan irregularities or damaged marks.
[16]	Dataset review of > 30 distinct sources (500 – 65000 OMR sheets) and five public sets (e.g., Afifi, Hussain, Rachchh, and Gopi).	Analyzes ten restriction categories (A–J): fiducial markers, template dependency, manual inputs, contrast requirements, etc.; maps 11/35 solutions as marker-dependent.	Findings: mean accuracy 98.9 %, mean error 1.08 %, avg time 2.76 s/sheet; recommends 4th generation OMR design (min param, flex layout, no marker, < 3 s/sheet, > 99 % accuracy).
[17]	Dataset: 40 synthetically deformed OMR images and 1035 real test sheets (240 dpi, 1984×2806 px) scanned via automatic feeder.	Detects ROI columns through geometric heuristics and applies quadrilateral transformation for alignment; implemented in C++ using AForge.NET.	Results: 100% success on synthetic set; 98.75% success on real 1035 images; average 1.25% failure due to merged or missing bubbles; efficient for batch OMR preprocessing.
[18]	Self-generated OMR forms.	Biogeography-Based Optimization for robust mark detection.	Claims accuracy improvement but no specific figures.
[19]	Institutional answer sheets (school-level).	Low-cost OMR software using standard scanners and image segmentation.	Focuses on affordability; lacks large-scale accuracy metrics.
[20]	Webcam frames under various lighting conditions.	Real-time OMR via webcam pipeline with noise handling.	Demonstrates latency reduction; lacks concrete speed/accuracy.
[21]	Hardware prototype with CCD sensor (3456px @ 5000fps).	FPGA-based OMR achieving real-time throughput.	Provides concrete timing specs; benchmarked vs. commercial OMRs.
[22]	University exam answer sheets.	Implements modified Multi-Connect Architecture (MMCA) for reading marks.	Hardware-oriented; speed evaluation not detailed.
[23]	Digital camera images of MCQ tests.	Automatic correction and grading via contour extraction.	Shows feasibility for handheld devices; no quantitative metrics.
[24]	Printed exam sheets (education domain).	Automated grading via adaptive thresholding and contour region detection.	Highlights fast marking; quantitative comparison missing.
[25]	64 students practicing programming via LMS (MCQs).	Experimental study comparing distributed vs. massed MCQ practice.	Includes statistical test ($p=0.031$) but unrelated to OMR processing.

REFERENCES

- [1] S. Maniar, J. Parmani, M. Bodke, and K. Saxena, "Generation and grading of arduous mcqs using nlp and omr detection using opencv," in *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 2021, pp. 1–7. DOI: 10.1109/ICCCNT51525.2021.9580089.
- [2] K. Tabassum and Z. Rahman, "Optical mark recognition with object detection and clustering," in *2024 6th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*, 2024, pp. 352–357. DOI: 10.1109/ICEEICT62016.2024.10534561.
- [3] M. Ali, S. Hammoud, and A. Esper, "Optical mark recognition techniques for multiple-choice tests: A comprehensive review," *IEEE Access*, vol. PP, pp. 1–1, Jan. 2025. DOI: 10.1109/ACCESS.2025.3600106.
- [4] T. Hieu, N. Phuong, and T. Thu, "Application of ai in grading new question types in the 2025 vietnamese high school graduation exam," *European Journal of Education Studies*, vol. 12, May 2025. DOI: 10.46827/ejes.v12i4.6007.
- [5] T. Jingyi, Y. K. Hooi, and O. K. Bin, "Image processing for enhanced omr answer matching precision," in *2021 International Conference on Computer Information Sciences (ICCOINS)*, 2021, pp. 322–327. DOI: 10.1109/ICCOINS49721.2021.9497172.
- [6] M. Afifi and K. Hussain, "The achievement of higher flexibility in multiple choice-based tests using image classification techniques," *International Journal on Document Analysis and Recognition (IJDAR)*, vol. 22, Jun. 2019. DOI: 10.1007/s10032-019-00322-3.
- [7] S. Tiwari and S. Sahu, "A novel approach for the detection of omr sheet tampering using encrypted qr code," in *2020 IEEE International Conference on Computational Intelligence and Computing Research*, 2020, pp. 1–5. DOI: 10.1109/ICCIC.2014.7238430.
- [8] R. Ahad, R. Toufiq, and S. Zaman, "Information retrieval from damage omr sheet by using image processing," Dec. 2019, pp. 153–156. DOI: 10.1109/ICECTE48615.2019.9303521.
- [9] B. Haskins, "Contrasting classifiers for software-based omr responses," in *2022 Pattern Recognition Association of South Africa and Robotics and Mechatronics International Conference (PRASA-RobMech)*, 2022, pp. 233–238. DOI: 10.1109/RoboMech.2015.7359528.
- [10] A. Saad, M. Mohamed, and E. Hossam, "Coverless image steganography based on optical mark recognition and machine learning," *IEEE Access*, vol. PP, pp. 1–1, Jan. 2021. DOI: 10.1109/ACCESS.2021.3050737.
- [11] J. Perez-Benedito, E. Aragon, J. Alriols, and L. Pejic, "Optical mark recognition in student continuous assessment," *Tecnologias del Aprendizaje, IEEE Revista Iberoamericana de*, vol. 9, pp. 133–138, Nov. 2024. DOI: 10.1109/RITA.2014.2363005.
- [12] G. R. I. Rasiq, A. A. Sefat, and M. Hasnain, "Mobile based mcq answer sheet analysis and evaluation application," Nov. 2023, pp. 144–147. DOI: 10.1109/SMART46866.2019.9117468.
- [13] H. Tjahyadi, S. Lukas, S. Albert, and D. Krisnadi, "Automated scoring of multiple-choice test using template matching technique," Sep. 2024, pp. 1–5, ISBN: 978-1-4503-6628-1. DOI: 10.1145/3284557.3284702.
- [14] D. N.C, S. Kaggere, and D. Ramegowda, "Automated parameter-less optical mark recognition: Proceedings of dal 2018," in Jan. 2019, pp. 185–195, ISBN: 978-981-13-2513-7. DOI: 10.1007/978-981-13-2514-4_16.
- [15] O. Espitia, A. Paez, Y. Mejia, M. Carrasco, and N. Gonzalez, "Optical mark recognition based on image processing techniques for the answer sheets of the colombian high-stakes tests," *IEEE*, vol. 1052, pp. 167–176, Oct. 2019.
- [16] E. Miguel de Elias, P. Tasinaffo, and R. Hirata, "Optical mark recognition: Advances, difficulties, and limitations," *SN Computer Science*, vol. 2, Sep. 2021. DOI: 10.1007/s42979-021-00760-z.
- [17] E. Miguel de Elias, P. Tasinaffo, and R. Junior, "Alignment, scale and skew correction for optical mark recognition documents based," Sep. 2019, pp. 26–31. DOI: 10.1109/WVC.2019.8876933.
- [18] H. Tjahyadi, S. Lukas, S. Albert, and D. Krisnadi, "A novel optical mark recognition technique based on biogeography based optimization," *IJITK*, vol. 5, no. 2, pp. 331–333, Dec. 2022.
- [19] V. Ware, N. M. Menon, P. Varute, and R. A. Dhannawat, "Cost effective optical mark recognition software for educational institutions," *International Journal of Advance Research, Ideas and Innovations in Technology*, vol. 5, pp. 1874–1877, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:189759603>.
- [20] H. Atasoy, E. Yildirim, Y. Kutlu, and K. Tohma, "Webcam based real-time robust optical mark recognition," vol. 9490, Nov. 2023, pp. 449–456, ISBN: 978-3-319-26535-3. DOI: 10.1007/978-3-319-26535-3_51.
- [21] S. Hussmann and P. W. Deng, "A high-speed optical mark reader hardware implementation at low cost using programmable logic," *Real-Time Imaging*, vol. 11, no. 1, pp. 19–30, 2020, ISSN: 1077-2014. DOI: <https://doi.org/10.1016/j.rti.2005.03.001>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1077201405000124>.
- [22] R. Hasan and E. Abdul Kareem, "An image processing oriented optical mark reader based on modify multi-connect architecture mmca," Mar. 2023.
- [23] F. Zampirolli, J. Quilici-Gonzalez, and R. Neves, "Automatic correction of multiple-choice tests using digital cameras and image processing," Apr. 2021.
- [24] D. Chai, "Automated marking of printed multiple choice answer sheets," Dec. 2019, pp. 145–149. DOI: 10.1109/TALE.2016.7851785.
- [25] L. Zhang, B. Li, Q. Zhang, and I.-H. Hsiao, "Does a distributed practice strategy for multiple choice questions help novices learn programming?" *International Journal of Emerging Technologies in Learning (iJET)*, vol. 15, no. 18, pp. 234–250, 2020. DOI: 10.3991/ijet.v15i18.10567. [Online]. Available: <https://online-journals.org/index.php/i-jet/article/view/10567>.